

Assignment – 3.5

Q - 1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

```
C:\Users\DHAVAL>ping 192.168.29.1

Pinging 192.168.29.1 with 32 bytes of data:
Reply from 192.168.29.1: bytes=32 time=4ms TTL=64
Reply from 192.168.29.1: bytes=32 time=5ms TTL=64
Reply from 192.168.29.1: bytes=32 time=7ms TTL=64
Reply from 192.168.29.1: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.29.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 7ms, Average = 5ms

C:\Users\DHAVAL>
```

Q - 2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
C:\Users\DHAVAL>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4827:400::]
over a maximum of 30 hops:

  1  *          *          *          Request timed out.
  2  *          *          *          Request timed out.
  3  *          *          13 ms    2405:203:400:100:172:31:3:216
  4  7 ms      4 ms      4 ms    2405:200:801:b00::ce4
  5  6 ms      5 ms      5 ms    2405:200:809:3168:61::5
  6  *          *          *          Request timed out.
  7  *          *          *          Request timed out.
  8  *          *          *          Request timed out.
  9  *          *          *          Request timed out.
 10  *          *          998 ms   2001:4860:4827:400::

Trace complete.

C:\Users\DHAVAL>
```

- **Command:** tracert www.youtube.com
- **Destination:** www.youtube.com
- **Number of hops:** 10 hops

Q - 3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```
C:\Users\DHAAVAL>ping 192.168.29.1

Pinging 192.168.29.1 with 32 bytes of data:
Reply from 192.168.29.117: Destination host unreachable.
Request timed out.
Reply from 192.168.29.117: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.29.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\DHAAVAL>|
```

Q - 4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```
Interface: 192.168.29.238 --- 0x12
Internet Address      Physical Address      Type
192.168.29.1          f0-ed-b8-4e-2a-93     dynamic
192.168.29.255        ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.29.117 --- 0x27
Internet Address      Physical Address      Type
192.168.29.1          f0-ed-b8-4e-2a-93     dynamic
192.168.29.255        ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\DHAVAL>
```

IP Address	Device / Gadget	Belongs to
192.168.29.238	Laptop	Me
192.168.29.117	Smartphone	Me

Q - 5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Hint: Include screenshots or command outputs if possible.

Introduction

The purpose of this activity was to set up a small Wi-Fi network and test whether different devices could communicate with each other. I used my laptop and my friend's phone, both connected to the same Wi-Fi network.

Network Setup

First, I connected my laptop to the home Wi-Fi network. Then, I connected my friend's phone to the same Wi-Fi network. I checked the phone's IP address and used it to test the connection from my laptop.

Ping Test

I opened Command Prompt on my laptop and entered the ping command followed by the phone's IP address. The laptop received replies from the phone, which showed that both devices could communicate successfully over the Wi-Fi network.

Traceroute Test

Next, I used the `tracert www.youtube.com` command. The command displayed the different network devices, or hops, between my laptop and YouTube. In my example, it took 12 hops to reach the destination. The number of hops can be different on different networks.

Disconnect Test

After that, I disconnected my friend's phone from the Wi-Fi. I tried the ping command again using the phone's previous IP address. This time, I received a "Request timed out" message because the phone was no longer connected to the network.

IP Address Table

Finally, I used the `arp -a` command to view IP and MAC addresses known to my laptop. I identified the devices such as the router, laptop, phone, and smart TV.

Conclusion

The activity helped me understand how devices communicate on a Wi-Fi network. The ping command was useful for checking connectivity, while `tracert` showed the path taken to reach a website. When the phone was disconnected, the ping failed, confirming that a device must be connected to the network to communicate with the laptop.